

How to Choose Data Stores & Tools in Microsoft Fabric?

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Who Am I?



Senior Cloud & Data Architect at Software Mind

Why this topic?

I've helped dozens of organizations choose the right data architecture in Microsoft Fabric
- and I want to share my thoughts with you

“

*Fabric has 6 data stores
and 5 ingestion tools.*

How to build solutions in MS Fabric?

”

What is important ...



For CFOs

How much will it cost?
How can we optimize
capacity?



For CDOs

How do we ensure
governance, security, and
compliance?



For Data Architects:

Which architecture is
scalable and
maintainable?

Agenda

01

Microsoft Fabric Big Picture

02

Ingestion Tools Overview & Comparison & When to Use Each Tool

03

Data Stores Overview & Comparison & When to Use Each Store

04

Decision Framework

05

Medallion Architecture in Fabric

06

Cost Optimization for CFOs

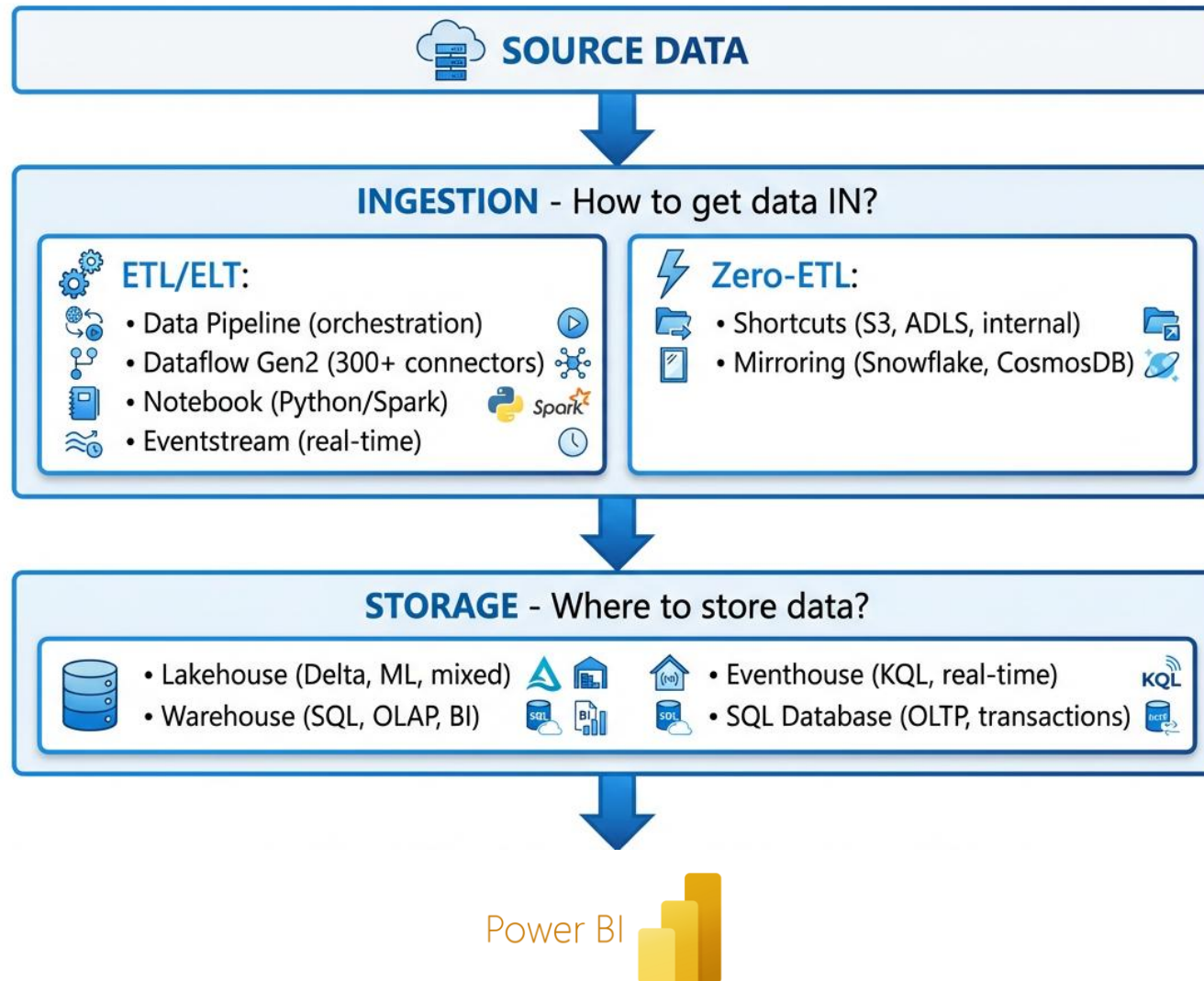
07

Governance for CDOs

08

Real Case Study

Microsoft Fabric - Big Picture



5 Ways to Ingest Data - Which to Choose?

Data Pipeline

Orchestration for large datasets, Azure sources, control flow

Dataflow Gen2

300+ connectors, low-code Power Query, on-premise support

Notebook

Python/Spark for complex transformations, API integration, validation

Eventstream

Real-time streaming from IoT Hub, Event Hub, Kafka

Shortcuts & Mirroring

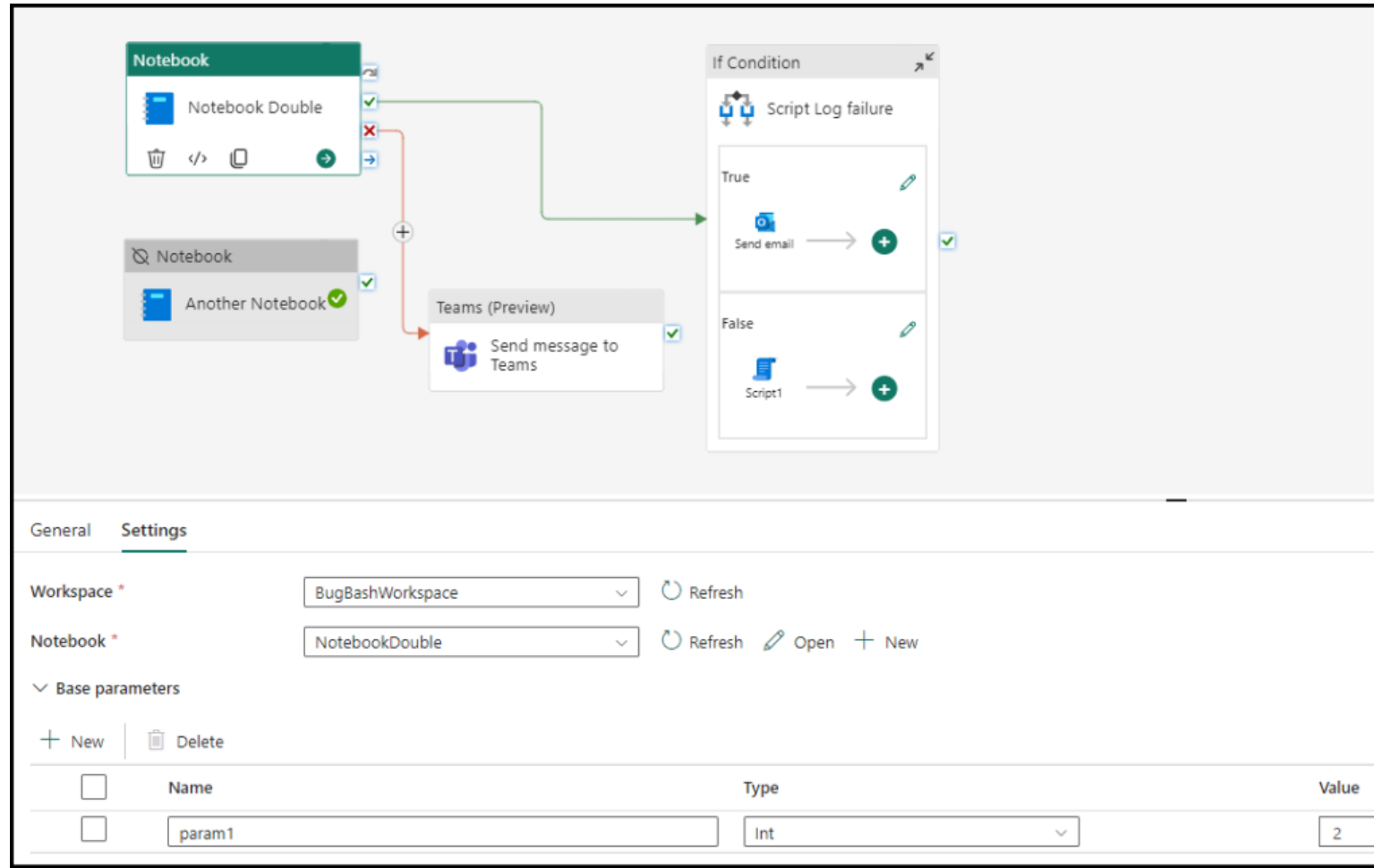
Zero-ETL: live link to S3/Snowflake (Shortcuts) or CDC replication (Mirroring)

Tools Comparison - MEGA TABLE

Criteria	Data Pipeline	Dataflow Gen2	Notebook	Eventstream	Shortcuts	Mirroring
Code?	No-code	Low-code	Code	No-code	No-code	No-code
Batch/RT?	Batch	Batch	Both	Real-time	N/A	Near RT
Skill	GUI	Power Query	Python/Spark		GUI	GUI
Complexity	Low	Medium	High	Medium	Low	Low
Large data?	No	Yes	Yes	Yes	No	No

- **Pipeline** best for orchestration, worst for on-premise
- **Dataflow** Gen2 has 300+ connectors but performance issues >100GB
- **Notebook** most flexible but requires Python skills
- **Eventstream** only for real-time, high CU consumption
- **Shortcuts** lowest cost (zero-copy), Mirroring for CDC

Data Pipeline



When to Use: Data Pipeline

✓ When to Use



Large datasets (TB+)



Orchestration & control flow



Cloud sources (Azure)



Triggering multiple actions



Scheduled batch processing

✗ When NOT to Use



On-premise data



Native transformations needed



Cross-workspace scenarios



Local file uploads



Real-time processing



Use for: TB+ datasets, orchestration, cloud sources, scheduled batch



Don't use for: on-premise data, native transformations, cross-workspace



Pro Tip: Orchestrate Dataflows and Notebooks for complex ETL workflows

Dataflow Gen2

Bronze1

Search

Trial: 54 days left

10

?

Home

Get data New semantic model Open notebook Manage OneLake data access (preview) Update all variables

A SQL analytics endpoint for SQL querying was created with this item.

Explorer

Search tables

Bronze1

Tables

incidents_sla.xlsx

ABC Number

ABC Priority

ABC Short description

ABC Configuration item

Created

Target Date

Completion Date

12L Business elapsed t...

12L Actual elapsed time

ABC Work Notes Log

ABC Assignment group

ABC Assigned to

Files

incidents_sla.xlsx

Table view

Showing 999 rows

	ABC Number	ABC Priority	ABC Short descri...	ABC Configurati...	Created	Target Date	Completion ...	12L Business ela...
1	INC14667065	P16	C\ Label: 02F16...	morstmsa01q	2/24/2024 6:02:3...	2/28/2024 3:30:0...	2/28/2024 7:29:1...	0
2	INC14637776	P40	TMS Appian Pro...	Translation Man...	2/14/2024 5:01:4...	2/21/2024 5:00:0...	2/26/2024 8:00:2...	158210
3	INC14667035	P16	C\ Label: F2EA7...	ca2stmsa03q	2/24/2024 5:47:2...	2/28/2024 3:30:0...	2/27/2024 11:43:...	0
4	INC14667066	P16	C\ Label: F2EA7...	ca2stmsa03c	2/24/2024 6:02:3...	2/28/2024 3:30:0...	2/27/2024 6:22:5...	0
5	INC14615363	P16	IQVIA Machine T...	Machine Translat...	2/6/2024 5:11:14...	2/8/2024 5:11:13...	2/6/2024 8:24:37...	11604
6	INC14640675	P16	IQVIA Machine T...	Machine Translat...	2/15/2024 3:12:1...	2/19/2024 3:12:1...	2/15/2024 3:42:5...	1835
7	INC14600813	P16	IQVIA Machine T...	Machine Translat...	2/1/2024 10:24:0...	2/5/2024 10:24:0...	2/1/2024 5:09:08...	24301
8	INC14611488	P16	IQVIA Translatio...	Machine Translat...	2/5/2024 6:11:21...	2/7/2024 6:11:20...	2/5/2024 7:32:34...	4874
9	INC14600789	P16	IQVIA Machine T...	Machine Translat...	2/1/2024 10:03:4...	2/5/2024 10:03:3...	2/1/2024 5:07:18...	7538
10	INC14633386	P16	IQVIA Machine T...	Machine Translat...	2/13/2024 12:22:...	2/15/2024 12:22:...	2/13/2024 6:09:0...	20794
11	INC14606543	P16	IQVIA Translatio...	Machine Translat...	2/2/2024 10:17:4...	2/6/2024 10:17:4...	2/6/2024 4:49:25...	1252
12	INC14640833	P16	IQVIA Machine T...	Machine Translat...	2/15/2024 4:07:5...	2/19/2024 4:07:5...	2/15/2024 10:28:...	1007
13	INC14623107	P16	I get a message ...	Machine Translat...	2/8/2024 6:06:53...	2/12/2024 6:06:5...	2/8/2024 9:31:12...	12260
14	INC14639712	P16	IQVIA Machine T...	Machine Translat...	2/15/2024 6:57:3...	2/19/2024 6:57:3...	2/22/2024 10:11:...	23278
15	INC14636416	P16	IQVIA Machine T...	Machine Translat...	2/14/2024 7:39:2...	2/16/2024 7:39:2...	2/15/2024 7:40:5...	86492
16	INC14653491	P16	IQVIA Machine T...	Machine Translat...	2/20/2024 5:56:2...	2/22/2024 5:56:2...	2/20/2024 6:26:0...	1776
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18	INC14613875	P16	IQVIA Translatio...	Machine Translat...	2/6/2024 8:41:44...	2/8/2024 8:33:46...	2/7/2024 8:14:04...	85218
19	INC14617708	P16	IQVIA Machine T...	Machine Translat...	2/7/2024 7:59:14...	2/9/2024 7:59:13...	2/7/2024 9:49:14...	6601

Succeeded (4 sec 467 ms)

Columns 12 Rows 999

When to Use: Dataflow Gen2

 **When to Use**

-  • 300+ connectors needed
-  • On-premise data access
-  • Low/no-code solution
-  • Power Query familiar team
-  • Extract, Transform, AND Load

 **When NOT to Use**

-  • Large datasets (>100GB)
-  • Data validation required
-  • Complex looping logic
-  • High performance needed
-  • Real-time processing

-  Use for: 300+ connectors, on-premise, low-code, Power Query teams
-  Don't use for: >100GB datasets, data validation, complex logic, high performance

 **Pro Tip: Staging consumes significant CU - monitor capacity usage carefully**

Notebook

The screenshot displays the Databricks Notebook interface. On the left, the 'Resources' sidebar shows a file explorer with a search bar and a tree view. The 'envtest (env)' folder is selected, containing 'train.tsv' and 'userdata1.parquet'. The main area features a code editor with three cells of Python code. The first cell imports 'Test' from 'builtin' and lists its members. The second cell imports 'matplotlib.pyplot' and 'mpimg' to load and display an image. The third cell imports 'pandas' to load CSV, JSON, and YAML files. The status bar at the bottom indicates 'Not connected', 'AutoSave: On', and 'Copilot completions: Off'.

Home Edit Run View

Comments Develop Share

PySpark (Python) Environment envtest Data Wrangler Copilot

All sources

Resources

Search files by name...

Built-in (builtin)

__pycache__

Test.py

envtest (env)

Dataset

train.tsv

userdata1.parquet

```
1 # Load python file
2 import builtin.Test as Test
3 # Now use the exported members from this module with identifier: `Test`
4 dir(Test)
```

[1] ✓ - Session ready in 14 sec 608 ms. Command executed in 4 sec 368 ms by PySpark (Python)

```
[[['jack', 34, '36'], ['Riti', 30, 'Delhi'], ['Aadi', 16, 'New York']]
['_builtins_',
 '__cached__',
 '__doc__',
 '__file__',
 '__loader__',
 '__name__',
 '__package__',
 '__spec__',
 'students']
```

```
1 # Note: NotebookUtils is only supported on runtime v1.2 and above. If you are using runtime v1.1, please use mssparkutils instead.
2 # Load image
3 import matplotlib.pyplot as plt
4 import matplotlib.image as mpimg
5 image = mpimg.imread(f"{notebookutils.nbResPath}/builtin/yourFile")
6 # Let the axes disappear
7 plt.axis('off')
8 # Plot image in the output
9 image_plot = plt.imshow(image)
```

PySpark (Python)

```
1 # Note: NotebookUtils is only supported on runtime v1.2 and above. If you are using runtime v1.1, please use mssparkutils instead.
2 import pandas as pd
3 # Load txt file
4 df_txt = pd.read_csv(f"{notebookutils.nbResPath}/builtin/yourFile.txt", sep="\s+")
5
6 # Load json file
7 df_json = pd.read_json(f"{notebookutils.nbResPath}/builtin/yourFile.json")
8
9 # Load yaml file
10 df_yaml = pd.read_csv(f"{notebookutils.nbResPath}/builtin/yourFile.yaml", sep="\s+")
11
```

Not connected AutoSave: On Copilot completions: Off

Selected Cell 1 of 3 cells

When to Use: Notebook

When to Use

-  API integration needed
-  Python client libraries
-  Data validation & quality
-  Code reuse & modularity
-  Large datasets with Spark

When NOT to Use

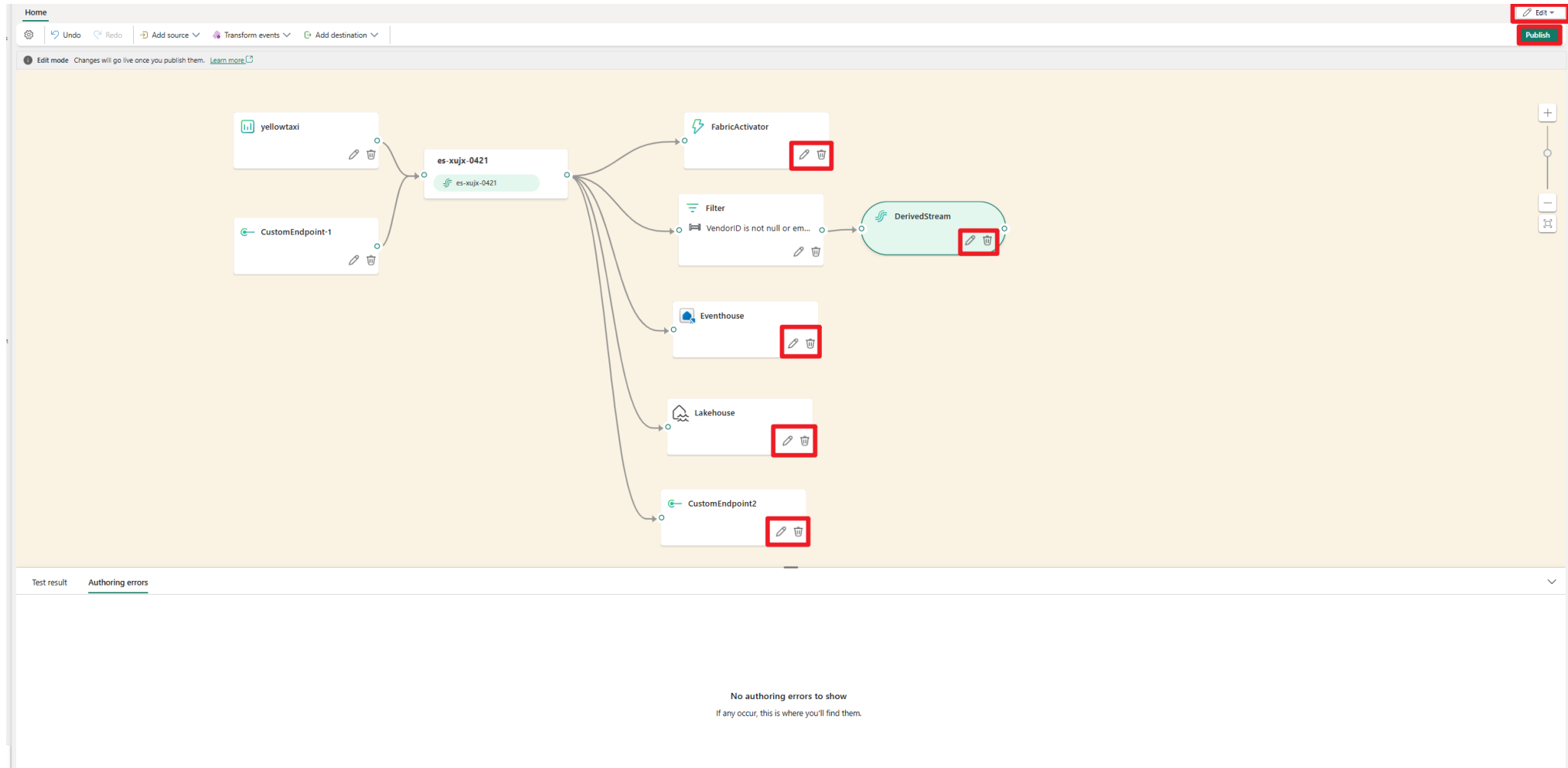
-  No Python skills in team
-  Simple ETL workflows
-  Low-code preference
-  Quick prototyping needed
-  Non-technical users

 Use for: API integration, Python libraries, data validation, code reuse, Spark

 Don't use for: no Python skills, simple ETL, low-code preference, non-technical users

 **Pro Tip: Perfect for complex transformations and ML workflows in Medallion Architecture**

When to Use: Eventstream



When to Use: Eventstream



When to Use:

- Real-time data streams (IoT, Event Hub, Kafka)
- Sub-second latency requirements
- Event-driven architectures
- Live dashboards and monitoring



When NOT to Use:

- Batch processing workloads
- Historical data only
- Cost-sensitive projects (high CU consumption)
- Complex transformations needed



Pro Tip: Eventstream is always-on compute - use only when real-time is truly required

When to Use: Shortcuts & Mirroring

Shortcuts - When to Use:

- Zero-ETL from S3, ADLS, Dataverse
- Read-only access to external data
- Cost optimization (lowest CU consumption)
- Data federation scenarios

Mirroring - When to Use:

- CDC from Azure SQL, Snowflake, Cosmos DB
- Near real-time replication
- Lower CU than full refresh Pipeline

 **Pro Tip: Shortcuts > Mirroring > Pipeline in terms of cost efficiency**

3 Data Stores in Fabric - Which to Choose?



Lakehouse

Delta Lake format, structured + unstructured, Spark transformations, ML/Data Science



Warehouse

T-SQL, strong RLS/CLS, multi-table transactions, BI reports, OLAP



Eventhouse

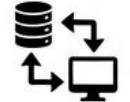
KQL Database, time-series, log analytics, real-time dashboards, IoT telemetry

When to Use: Lakehouse

✓ When to Use

-  • ML & Data Science
-  • Mixed data (structured + files)
-  • Medallion Architecture
-  • Delta Lake format
-  • Exploratory analytics

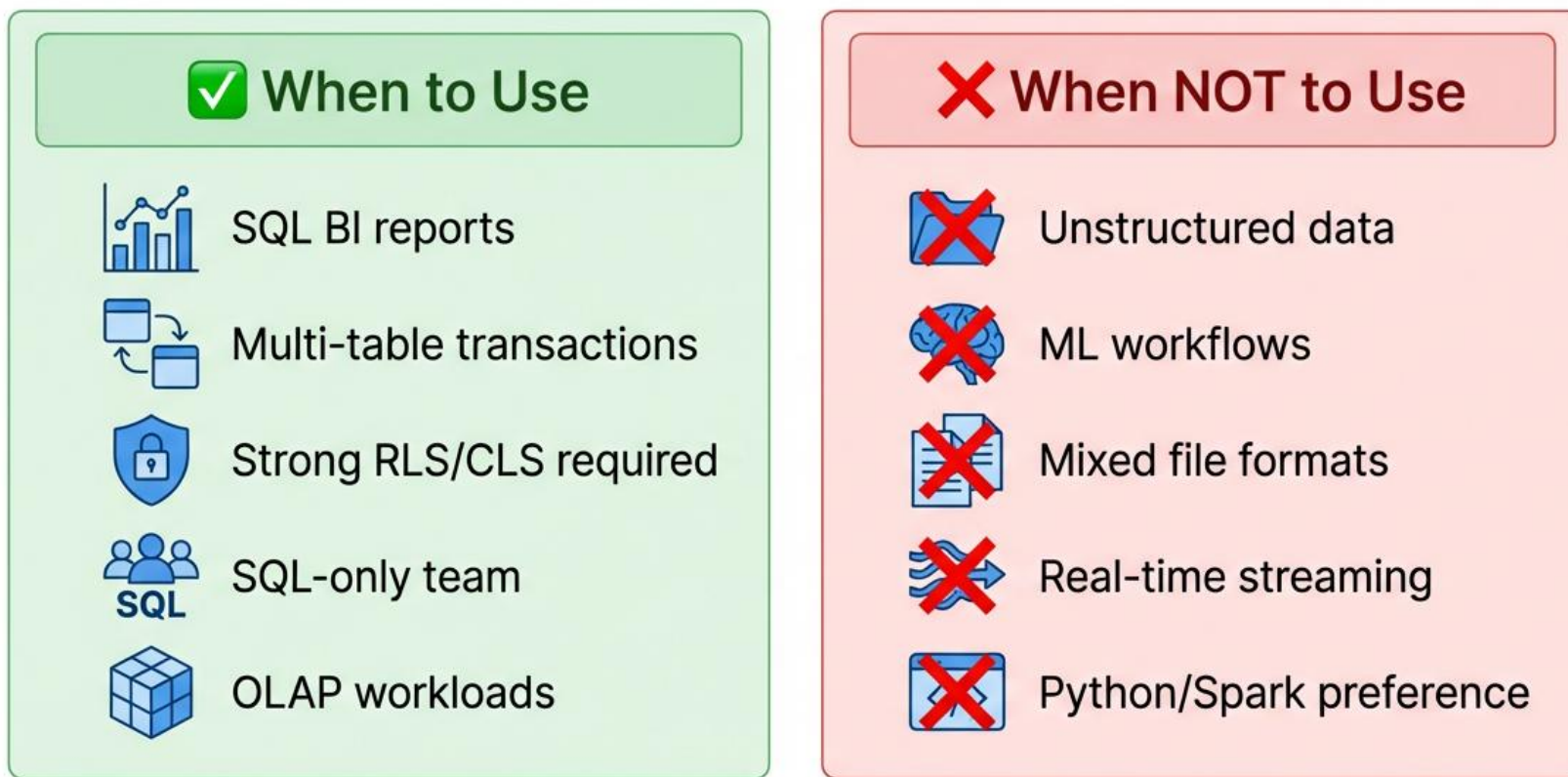
✗ When NOT to Use

-  • Simple SQL reports
-  • Multi-table transactions
-  • Strong RLS/CLS needed
-  • SQL-only team
-  • OLTP workloads

- ✓ Use for: ML/Data Science, mixed data, Medallion Architecture, Delta Lake, exploratory analytics
- ✗ Don't use for: simple SQL reports, multi-table transactions, strong RLS/CLS, SQL-only teams, OLTP

💡 **Pro Tip: Ideal for Bronze/Silver layers with Spark transformations**

When to Use: Warehouse

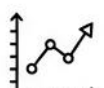


✓ Use for: SQL BI reports, multi-table transactions, strong RLS/CLS, SQL-only teams, OLAP

✗ Don't use for: unstructured data, ML workflows, mixed file formats, real-time streaming, Python/Spark preference

💡 **Pro Tip: Perfect for Gold layer with strong governance needs**

When to Use: Eventhouse

✓ When to Use	✗ When NOT to Use
 Real-time data streams	 Batch analytics
 IoT & telemetry	 SQL workloads
 Log analytics	 Complex joins
 Time-series data	 Cost-sensitive projects
 KQL-familiar team	 Historical data only

✓ Use for: real-time streams, IoT/telemetry, log analytics, time-series, KQL-familiar teams

✗ Don't use for: batch analytics, SQL workloads, complex joins, cost-sensitive projects, historical data only

💡 **Pro Tip: Excels at real-time dashboards but has higher CU consumption - use wisely**

Data Stores Comparison - MEGA TABLE

Criteria	Lakehouse	Warehouse	Eventhouse
Data type	Mixed (structured + files)	Structured	Event/logs
Skill	Spark/SQL	SQL	KQL
Scale	PB	PB	PB/s
Latency	s-min	s	ms
Multi-table TX	✗	✓	✗
ML/Advanced	✓	✗	✗
Real-time	✗	✗	✓
Shortcuts	✓	✓	✓
RLS/CLS	⚠	✓	✓

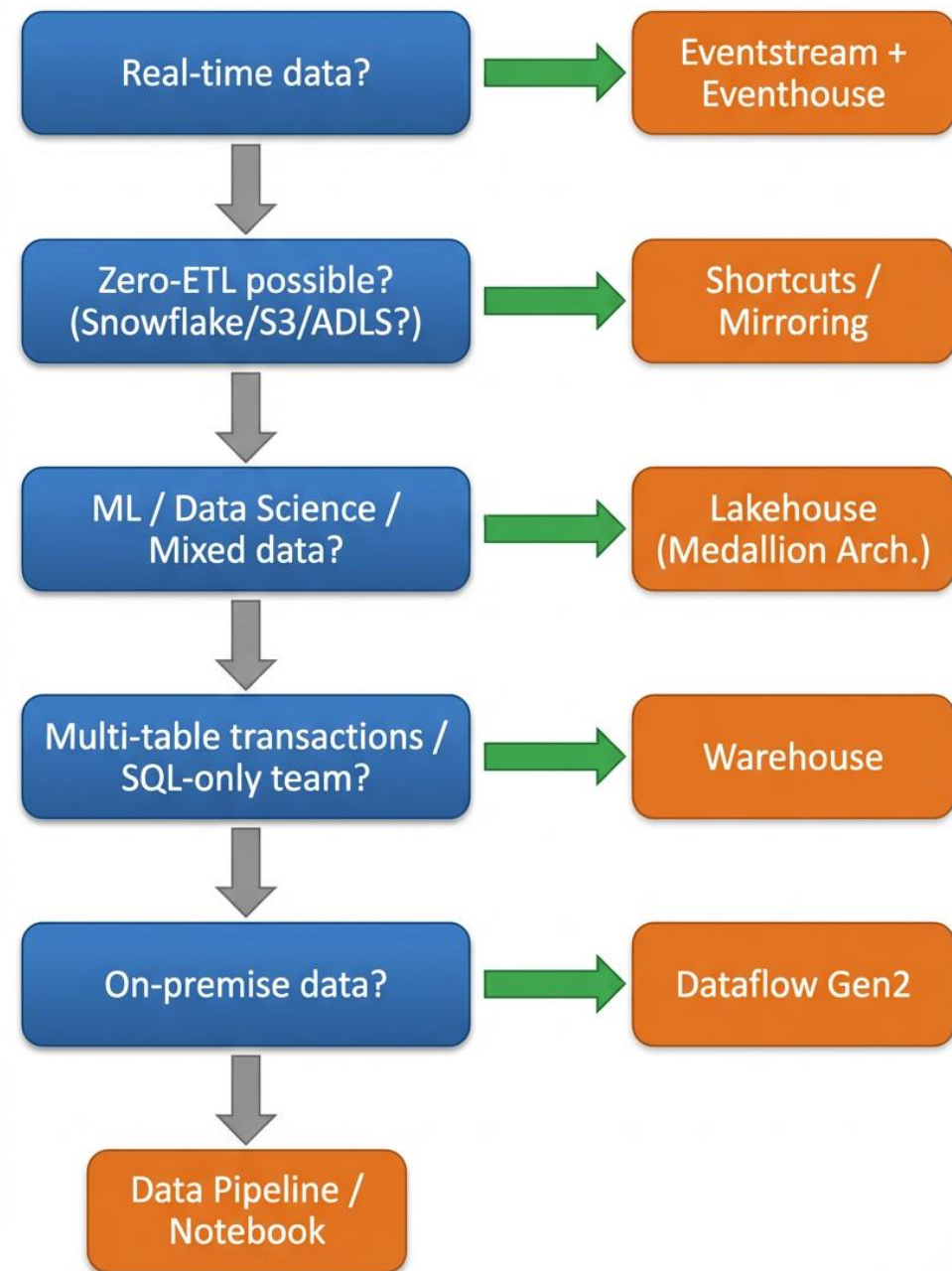
Lakehouse: Delta Lake, limited RLS, best for ML/Data Science

Warehouse: T-SQL, strong RLS/CLS, best for BI reports

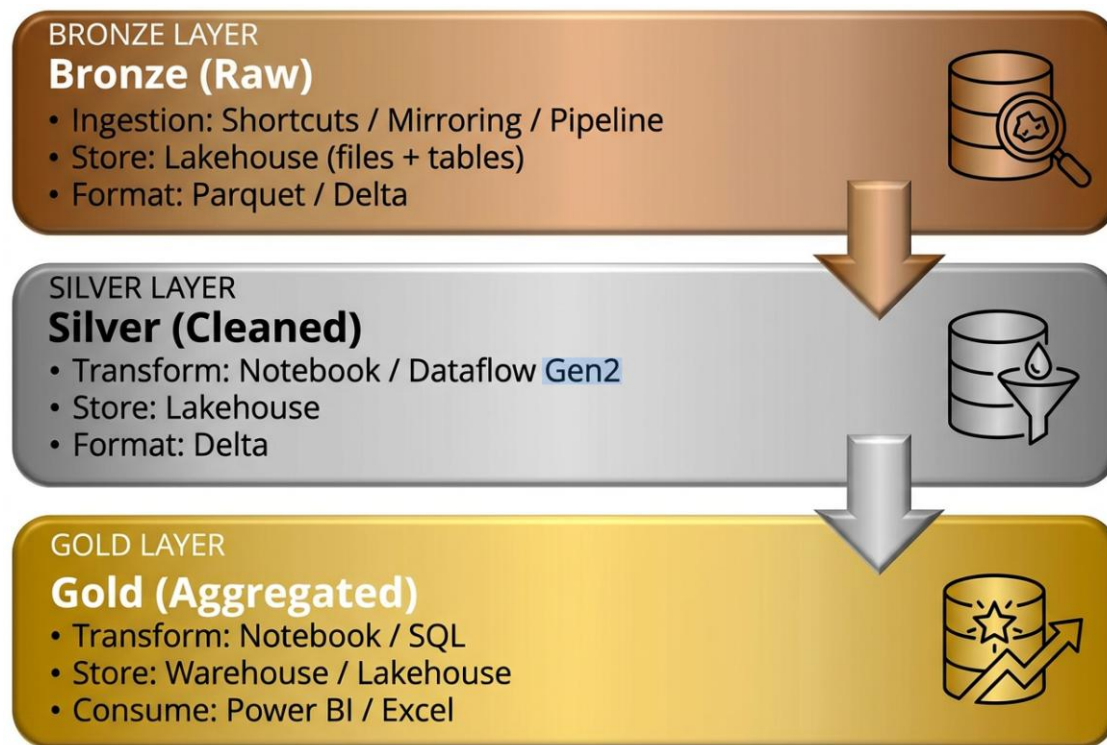
Eventhouse: KQL, real-time, high CU consumption

Decision Framework

1. Start with data velocity: batch vs real-time,
2. Consider: source type, transformation complexity, team skills,
3. End with: governance needs and cost sensitivity,
4. This flowchart guides you to the right tool + store combination



Medallion Architecture in Fabric



Bronze: Lakehouse + Shortcuts (raw data, zero-copy)

Silver: Lakehouse + Notebooks (cleansed, validated, PySpark)

Gold: Warehouse (BI reports, strong RLS) or Lakehouse (ML models)

Microsoft Purview + Fabric Integration



Data Catalog & Discovery

- Auto-scan Lakehouse, Warehouse, Semantic Models
- Centralized metadata repository
- Self-service data discovery without IT

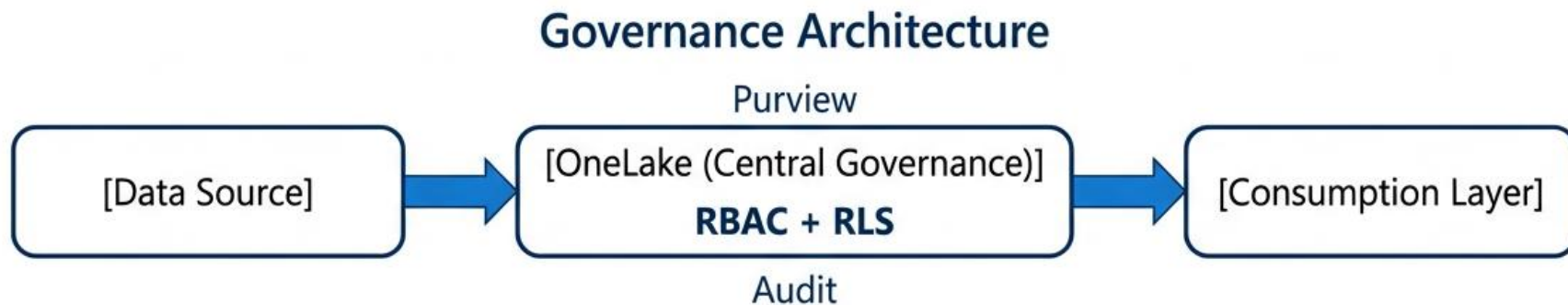
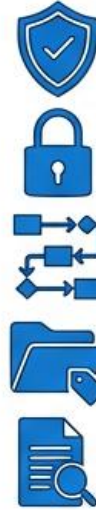


Data Lineage

- End-to-end lineage: Source → Pipeline → Lakehouse → Warehouse → Power BI
- Impact analysis for changes
- Root cause analysis for data quality

How to speak with CDOs?

- ✓ OneLake Security - RBAC on workspace/item level
- ✓ RLS/CLS - Warehouse/Eventhouse (Lakehouse limited)
- ✓ Data Lineage - automatic in Fabric
- ✓ Purview Integration - data catalog, sensitivity labels
- ✓ Audit Logs - all operations in tenant



How to speak with CFOs?

Pay-as-you-go

Flexible model



- Monthly payment based on actual usage
- Scale up/down based on current needs
- Option to pause capacity to reduce costs
- Ideal for variable or uncertain workloads

Like paying for electricity - you only pay for what you use.

Reserved (1 Year)

Fixed annual rate



- Pre-purchase committed capacity
- Lower cost with annual savings
- Predictable budgeting
- Best for stable, long-term workloads

Like a subscription - pay upfront and save.

How to speak with CFOs

Autoscale for Spark



Dynamic Spark Workloads

- Dedicated serverless Spark compute
- Auto-scales without impacting capacity
- No resource contention
- Pay only for active Spark compute
- Standard PAYG CU-hour rates

Opt-in model

Capacity Overage



Avoid Throttling

- Pay for excess consumption
- Uninterrupted operations
- Set 24-hour overage limits
- Checked every 5 minutes
- PAYG pricing (no reservation discount)

Mission-critical workloads

Dedicated Capacity



Reserved Resources

- Fixed capacity reserved for you
- Predictable performance
- Full data integration
- Monthly or annual plans
- Discounted rates

Enterprise model

How to speak with CFOs?

Microsoft Fabric Capacity Pricing

SKU	Capacity Units (CU)	Pay-as-you-go	Reserved (1 year)
F2	2	\$321.20 /month	\$191.00 /month
F4	4	\$642.40 /month	\$382.00 /month
F8	8	\$1,284.80 /month	\$764.00 /month
F16	16	\$2,569.60 /month	\$1,556.00 /month

Storage Pricing

OneLake Storage

Storage Type	Price per GB/month
OneLake storage	\$0.024/GB
OneLake BCDR storage	\$0.046/GB
OneLake cache*	\$0.26/GB

*OneLake cache is billed for KQL cache storage and Data Activator data retained.

SQL Storage

Storage Type	Price per GB/month
SQL Storage (Data Warehouse)	\$0.022/GB
SQL Log Storage	\$0.12/GB

Mirroring: Free Storage for Replicas

Free storage based on purchased capacity SKU (only for Mirroring replica data)

Capacity SKU	Free Mirroring Storage (TB)
F2	2 TB
F4	4 TB
F8	8 TB
F16	16 TB
F32	32 TB
F64 / P1	64 TB
F128 / P2	128 TB
F256 / P3	256 TB
F512 / P4	512 TB

How to speak with CFOs?

Fabric Built-in Cost Control Tools



Capacity Admin Portal

Real-time CU monitoring dashboard



Throttling & Smoothing

Automatic limits on overconsumption



Overage Limits

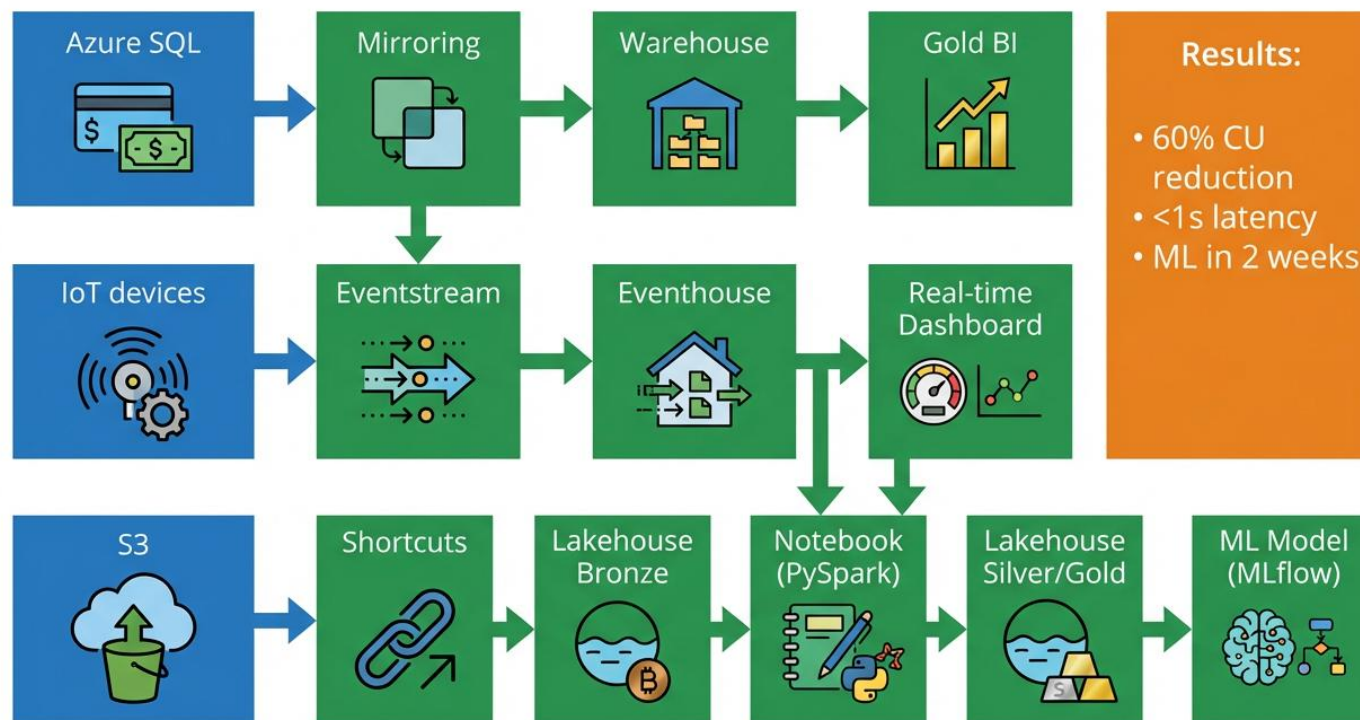
Set max spend per 24 hours



Autoscale Limits

Admin controls max CU for Spark

Case Study: E-commerce Platform



Azure SQL → Mirroring → Warehouse → BI reports

IoT devices → Eventstream → Eventhouse → Real-time dashboards

S3 → Shortcuts → Lakehouse Bronze

All sources → Notebooks → Lakehouse Silver/Gold → ML models

Results: 60% CU reduction, <1s latency, ML in 2 weeks

Decision Cheat Sheet - Take This Home

	Scenario	Recommended Tool	Recommended Store
	Real-time IoT	Eventstream	Eventhouse
	SQL BI reports	Mirroring / Pipeline	Warehouse
	ML / Data Science	Notebook	Lakehouse
	On-premise data	Dataflow Gen2	Lakehouse/Warehouse
	Zero-ETL (S3/Snowflake)	Shortcuts / Mirroring	Lakehouse/Warehouse
	Cost optimization	Shortcuts > Pipeline	–

Real-time IoT → Eventstream + Eventhouse

SQL BI → Mirroring/Pipeline + Warehouse

ML/Data Science → Notebook + Lakehouse

On-premise → Dataflow Gen2 + Lakehouse/Warehouse

Zero-ETL → Shortcuts/Mirroring + Lakehouse/Warehouse

Cost optimization → Shortcuts over Pipeline



Thank You!
Questions?